Interpretation of Descriptive Analysis and Percentiles

Academic Percentages and Salary

This assignment explains the information that can be interpreted from the supplied statistical table. The serial-number column (sl_no) is excluded because it is an identifier rather than a meaningful measurement. The interpretation below is based only on the displayed values.

Data set file name: Placement.csv

1. Understanding the Rows in the Table

- **Mean:** The arithmetic average of all values in a column. It is calculated by adding the observations and dividing by the number of observations. Extreme values can affect the mean.
- **Median:** The middle value after the observations are arranged in order. Half the observations are at or below it and half are at or above it, allowing for ties.
- **Mode:** The value that occurs most frequently. A dataset may have more than one mode.
- **Q1: 25%:** The first quartile, or 25th percentile. Approximately 25% of observations are at or below this value, and approximately 75% are at or above it.
- **Q2: 50%:** The second quartile, or 50th percentile. This is the median.
- **Q3: 75%:** The third quartile, or 75th percentile. Approximately 75% of observations are at or below this value, and approximately 25% are at or above it.
- **99%:** The 99th percentile. Approximately 99% of observations are at or below this value, with approximately 1% above it, subject to the percentile calculation method and ties.
- **Q4: 100%:** The 100th percentile, normally the maximum observed value in the column.

2. Values Reported in the Table

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Q4:100%	215.0	89.4	97.7	91.0	98.0	77.89	940000.0

3. Step-by-Step Interpretation of Each Variable

3.1 SSC Percentage (ssc_p)

- Mean = 67.30%. The average SSC percentage is approximately 67.30%.
- Median / Q2 = 67%. The middle value is 67%.
- Mode = 62%. The most frequently occurring SSC percentage is 62%.
- Q1 = 60.6%. Approximately one-quarter of the SSC observations are at or below 60.6%.
- Q3 = 75.7%. Approximately three-quarters of the observations are at or below 75.7%; the top quarter is at or above this point.
- 99th percentile = 87%. About 99% of the observations are at or below 87%, with roughly 1% above it, depending on ties and calculation method.
- Maximum (100th percentile) = 89.4%. The highest SSC percentage shown by the table is 89.4%.
- Overall interpretation: The middle 50% of SSC percentages lie approximately between 60.6% and 75.7%. The mean is close to the median, while the mode is lower.

3.2 HSC Percentage (hsc_p)

- Mean = 66.33%. The average HSC percentage is approximately 66.33%.
- Median / Q2 = 65%. The middle HSC percentage is 65%.
- Mode = 63%. The most frequently occurring HSC percentage is 63%.
- Q1 = 60.9%. Approximately 25% of HSC observations are at or below 60.9%.
- Q3 = 73%. Approximately 75% of observations are at or below 73%, and the upper quarter is at or above this value.
- 99th percentile = 91.86%. Approximately 99% of HSC observations are at or below 91.86%.
- Maximum = 97.7%. The highest HSC percentage in the table is 97.7%.
- Overall interpretation: The middle 50% of HSC percentages fall approximately between 60.9% and 73%. The mean is slightly above the median and mode.

3.3 Degree Percentage (degree_p)

- Mean = 66.37%. The average degree percentage is approximately 66.37%.
- Median / Q2 = 66%. The middle degree percentage is 66%.
- Mode = 65%. The most frequently occurring degree percentage is 65%.
- Q1 = 61%. Approximately 25% of degree percentages are at or below 61%.
- Q3 = 72%. Approximately 75% of observations are at or below 72%.
- 99th percentile = 83.86%. Approximately 99% of degree percentages are at or below 83.86%.
- Maximum = 91%. The highest degree percentage shown is 91%.
- Overall interpretation: The middle 50% of degree percentages lie approximately between 61% and 72%. Mean, median, and mode are close to one another, so their central values are similar.

3.4 Employability Test Percentage (etest_p)

- Mean = 72.10%. The average employability-test percentage is approximately 72.10%, the highest mean among the academic percentage variables in the table.
- Median / Q2 = 71%. The middle score is 71%.
- Mode = 60%. The most frequently occurring score is 60%.
- Q1 = 60%. Approximately 25% of test scores are at or below 60%.
- Q3 = 83.5%. Approximately 75% of test scores are at or below 83.5%; the top quarter is at or above this value.
- 99th percentile = 97%. Approximately 99% of test scores are at or below 97%.
- Maximum = 98%. The highest employability-test percentage shown is 98%.

- Overall interpretation: The middle 50% of scores fall approximately between 60% and 83.5%. The mean and median are higher than the mode, indicating that the most common score is lower than the average and middle score.

3.5 MBA Percentage (mba_p)

- Mean = 62.28%. The average MBA percentage is approximately 62.28%, the lowest mean among the academic percentage variables shown.
- Median / Q2 = 62%. The middle MBA percentage is 62%.
- Mode = 56.7%. The most frequently occurring MBA percentage is 56.7%.
- Q1 = 57.945%. Approximately 25% of MBA percentages are at or below 57.945%.
- Q3 = 66.255%. Approximately 75% of MBA percentages are at or below 66.255%.
- 99th percentile = 76.1142%. Approximately 99% of MBA percentages are at or below 76.1142%.
- Maximum = 77.89%. The highest MBA percentage shown is 77.89%.
- Overall interpretation: The middle 50% of MBA percentages lie approximately between 57.945% and 66.255%. The mean and median are close, while the mode is lower.

3.6 Salary

- Mean = 288,655.41. The average salary is approximately 288,655.41 in the salary units used in the original dataset.
- Median / Q2 = 265,000. The middle salary is 265,000.
- Mode = 300,000. The most frequently occurring salary is 300,000.
- Q1 = 240,000. Approximately 25% of salary observations are at or below 240,000.
- Q3 = 300,000. Approximately 75% of salary observations are at or below 300,000, and approximately 25% are at or above this value.
- 99th percentile = NaN. The table does not provide a usable 99th-percentile salary value. NaN usually means “Not a Number” and can indicate a missing or undefined value; the table alone does not explain why it appears.
- Maximum = 940,000. The highest salary shown is 940,000.
- Overall interpretation: The middle 50% of salaries lie approximately between 240,000 and 300,000. The mean is above the median, which may indicate that some higher salaries pull the average upward. The currency and time period are not specified, so neither should be assumed.

4. Comparisons and Key Findings

- Highest mean among academic percentages: Employability test, approximately 72.10%.
- Lowest mean among academic percentages: MBA, approximately 62.28%.
- SSC and degree means are similar: approximately 67.30% and 66.37%.
- The mean and median are close for SSC, degree, and MBA percentages, indicating similar average and middle values. This alone does not prove that the distributions are symmetric.
- For all five academic percentage variables, the mode is lower than the mean. The most common score is therefore lower than the average score in each column.
- Salary mean (about 288,655.41) exceeds its median (265,000), and the maximum is 940,000. Some high salary observations may be pulling the mean upward, but the table alone cannot confirm the exact distribution shape.
- For each variable, Q2 equals the median, as expected because both represent the 50th percentile.
- The 99th-percentile salary is shown as NaN. This value should be reported as unavailable rather than interpreted as zero.

5. Conclusion

The table provides a useful statistical summary of five academic percentage variables and salary. The mean describes the average, the median or Q2 describes the middle value, and the mode identifies the most frequent value. Q1 and Q3 show the boundaries of the middle 50% of observations, the 99th percentile indicates a value near the upper end of the distribution, and the 100th percentile gives the maximum. The employability-test percentage has the highest academic mean and the widest middle-50% spread, while MBA percentage has the lowest academic mean and the narrowest middle-50% spread. Salary has a mean above its median and a maximum of 940,000, while its 99th-percentile value is unavailable in the table. These results describe the data but should be combined with further analysis for a fuller understanding.